

Manuscripts considered — pdac-recurrent-cldn18-2-kras-g12r-tp53 -y220c-p25m

Master inventory: every paper reviewed in this case

Generated 2026-08-12

Manuscripts considered — pdac-recurrent-cldn18-2-kras-g12r-tp5

Master inventory: 32 clinical (24 included, 8 considered & excluded) + 46 pre-clinical (39 included, 7 considered & excluded) + 47 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

O'Reilly/Wolpin (2026) — N Engl J Med

Reference: [PMID 42223072](#) · Type: clinical · Inclusion: included

- **Intervention:** daraxonrasib (RMC-6236, pan-RAS(ON) multi-selective inhibitor)
- **Indication / model:** Previously treated metastatic PDAC (91.8% RAS G12-mutant) (2L); Previously treated mPDAC; dual primary endpoints in the RAS G12 subpopulation, which contains this patient's G12R. Overall population also took G13/Q61 and RAS-WT.
- **Design:** Randomized open-label phase 3 (RASolute 302, NCT06625320)
- **n:** 500
- **Effect size:** 13.2 months — median OS, RAS G12 population
- **Variance:** HR CIs not in abstract; $P < 0.001$ for OS and PFS comparisons; $p < 0.001$
- **Toxicities:** any AE (n=248, 100%); any AE $G \geq 3$ (n=248, 61.8%); TRAE leading to discontinuation (n=248, 1.2%)
- **Case fit:** strong
- **Notes:** Pivotal randomized win in exactly this population; FDA submission announced. Per-term grade ≥ 3 AE table sits in the NEJM full text; abstract-level union rates captured here and CT.gov results were not posted at review (checked August 2026). Eligibility required metastatic disease, so restaging must show the recurrence is metastatic for this estimand to transfer cleanly.

Wolpin/Hong (2026) — N Engl J Med

Reference: [PMID 42090791](#) · Type: clinical · Inclusion: included

- **Intervention:** daraxonrasib (RMC-6236, pan-RAS(ON) multi-selective inhibitor)
- **Indication / model:** Previously treated advanced RAS-mutated PDAC (phase 1/2 solid-tumor study, PDAC subset) (2L+); 168 previously treated RAS-mutant PDAC patients dosed at ≤ 300 mg; key efficacy subgroup is 26 patients with RAS G12 tumors on second-line 300 mg. G12R qualifies alongside the other G12 alleles.
- **Design:** Phase 1/2 dose escalation/optimization, single-arm (RMC-6236-001, NCT05379985)
- **n:** 168
- **Effect size:** 35 % — ORR, 2L RAS G12 at 300 mg
- **Variance:** 95% CI 17–56
- **Toxicities:** any TRAE (n=168, 96%); any TRAE $G \geq 3$ (n=168, 30%); rash (n=168); diarrhea (n=168); nausea (n=168); stomatitis/mucositis (n=168); vomiting (n=168); fatigue (n=168)
- **Case fit:** strong
- **Notes:** The single-arm numbers behind RASolute 302; the 2L G12 subgroup is small (n=26), so the 35% ORR carries a wide CI. No G12R-selective inhibitor exists in the clinic; the pan-RAS(ON) mechanism is the allele-agnostic route for G12R, and the trial is still recruiting.

Dumbrava/Schram (2026) — N Engl J Med

Reference: [PMID 41740031](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** rezatapopt (PC14586, p53 Y220C reactivator)
- **Indication / model:** Locally advanced/metastatic solid tumors with TP53 Y220C (heavily pretreated) (2L+); 77 patients across 8 dose levels. Every responder had a KRAS wild-type tumor; ORR 30% in KRAS-WT at ≥ 1150 mg vs 20% overall.
- **Design:** Phase 1 single-arm dose escalation/optimization (PYNACLE, NCT04585750)
- **n:** 77
- **Effect size:** 20 % — ORR (all patients)
- **Toxicities:** nausea (n=77, 58%); vomiting (n=77, 44%); blood creatinine increased (n=77, 39%); fatigue (n=77, 39%); anemia (n=77, 36%); anemia $G \geq 3$ (n=77, 16%); TRAE leading to discontinuation (2/77, 3%)
- **Case fit:** partial
- **Notes:** Proof of concept for p53 reactivation, but the fine print cuts against this case twice: the registrational phase 2 monotherapy cohort excludes any known KRAS SNV (G12R blocks it), and in phase 1 every responder was KRAS wild-type — so monotherapy benefit in a KRAS G12R tumor is unproven at best. The pembrolizumab combination arm carries no KRAS exclusion but leans on checkpoint biology this MSS/TMB-low tumor lacks.

—/— (2026)

Reference: — · **Type:** clinical · **Inclusion:** excluded — allele mismatch — zoldonrasib is KRAS G12D(ON)-selective and its trials require documented G12D; a G12R tumor fails eligibility outright

- **Intervention:** zoldonrasib (RMC-9805, KRAS G12D(ON)-selective inhibitor)
- **Notes:** Excluded: allele mismatch — zoldonrasib is KRAS G12D(ON)-selective and its trials require documented G12D; a G12R tumor fails eligibility outright — Reviewed to confirm the allele gate; its clinical data (G12D PDAC/NSCLC cohorts, conference-level) were not extracted because the drug cannot apply to this patient. Same sponsor's daraxonrasib is the allele-appropriate route.

—/— (2026)

Reference: — · **Type:** clinical · **Inclusion:** excluded — no published clinical evidence — first-in-human dose escalation (NCT07458347) opened 2026 with any-KRAS-mutation eligibility

- **Intervention:** KST-6051 (oral pan-KRAS inhibitor)
- **Notes:** Excluded: no published clinical evidence — first-in-human dose escalation (NCT07458347) opened 2026 with any-KRAS-mutation eligibility — Trial-eligibility route only; nothing to extract.

Burge/Hobbs (2026) — Cancer Research

Reference: [PMID 41556816](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** KRAS G12R allele biology (target-feature mechanism; not a single agent)
- **Indication / model:** KrasLSL-G12R/+; Trp53LSL-R172H/+; p48Cre-ER GEMM; murine and human PDAC cell lines; PDX models; human PDAC tissue
- **Design:** G12R produces less nuclear ERK/MAPK translocation and therefore weaker transcriptional output than

G12D, and direct KRAS-driven PI3K activation is what the mouse model needs for efficient tumor initiation.

- **Effect size:** strong — Only about 10% of KrasG12R;Trp53R172H mice formed pancreatic tumors within a year, against rapid multifocal disease in G12D animals. Both alleles switched on the same pancreas-specific transcriptional network, but G12R did so weakly. Human G12R tumors carried less collagen deposition and less metastatic liver invasion.
- **Variance:** vs KrasG12D GEMM and G12D-mutant human lines/PDX; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** The mouse p53 allele is R172H, so the co-mutation is analogous to this patient's Y220C rather than matched. Reduced collagen in human G12R tumors cuts against the desmoplastic-barrier story for this allele, but that part is an association in tissue, not an intervention.

Grimont/Chandwani (2026) — Developmental Cell

Reference:PMID 42480541 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** KRAS G12R allele biology (target-feature mechanism; not a single agent)
- **Indication / model:** Ptf1a-tdTomato lineage-tracing mice expressing different KRAS mutants, with genetic, pharmacologic and inflammatory perturbation in vivo
- **Design:** G12R and G12V fail to raise EGFR, AKT and RAC1/VAV1 signalling enough to license acinar-to-ductal lineage reversion and the accompanying enhancer reprogramming at Pou2f3 and Vav1.
- **Effect size:** strong — G12D co-opted pancreatic injury to drive dedifferentiation and tumor initiation, while G12R and G12V could not sustain the dedifferentiated transcriptional or chromatin programs. Constitutive AKT activation was sufficient to rescue G12R tumorigenic potential in vivo.
- **Variance:** vs KRAS G12D mice under the same perturbations; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Tumor-initiation biology in mice, not treatment of established disease. Its therapeutic reading is indirect: an allele that leans on AKT to get started may not behave like G12D under MAPK-directed pressure.

Aronchik/Singh (2026) — Nature Medicine

Reference:PMID 42581230 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236, pan-RAS(ON) multi-selective inhibitor) — acquired resistance
- **Indication / model:** human and murine PDAC preclinical models, anchored to paired pretreatment/end-of-treatment ctDNA from 44 phase 1/2 patients
- **Design:** Escape restores RAS pathway signalling rather than altering the drug site: mutant KRAS amplification, MYC amplification, and receptor tyrosine kinase upregulation.
- **n:** 44 paired ctDNA samples; model arms not enumerated in the abstract
- **Effect size:** moderate — RAS-pathway alterations emerged in 26 of 44 patients and mutant KRAS amplification in 16 of 44, with no secondary KRAS mutations at all, unlike the G12C(OFF) resistance pattern. Pairing daraxonrasib with DNA-damage-response agents, RTK inhibitors, or zoldonrasib averted resistance in models.
- **Variance:** translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Cohort is RAS-mutant PDAC broadly, with no G12R breakdown. The zoldonrasib pairing is G12D-directed and does not transfer to this tumor.

Dorbin/McFall (2026) — bioRxiv (preprint)

Reference: [PMID 42465401](#) · Type: preclinical · Inclusion: included

- **Intervention:** daraxonrasib (RMC-6236) — allele-specific resistance in KRAS G12R
- **Indication / model:** KRAS G12R vs G12D PDAC cell lines; 3D-bioprinted patient-derived toroids; one G12R patient's tumor tissue
- **Design:** G12R cannot transactivate wild-type RAS, so daraxonrasib-adapted G12R cells lean on EGFR-driven wild-type RAS-GTP, while G12D cells retain mutant KRAS-GTP signalling with reduced cyclophilin A, the partner daraxonrasib needs.
- **Effect size:** moderate — The shift to EGFR/RAS-WT in resistant G12R cells conferred trametinib sensitivity, which the bioprinted toroid model reproduced. A G12R patient who progressed after 10 months on daraxonrasib showed intratumoral EGFR/RAS-WT activation and got roughly 5 months of control on sixth-line trametinib-based therapy.
- **Variance:** vs KRAS G12D lines under matched daraxonrasib pressure; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** bioRxiv preprint, not peer reviewed, with a single-patient clinical anchor. Read it as a hypothesis about what follows daraxonrasib in a G12R tumor, not as established resistance biology.

McAndrews/Kalluri (2026) — The EMBO Journal

Reference: [PMID 42436354](#) · Type: preclinical · Inclusion: included

- **Intervention:** daraxonrasib / zoldonrasib resistance via CDK8-driven microenvironment remodeling
- **Indication / model:** PDAC mouse models profiled by spatial transcriptomics, scRNA-seq and CODEX; PDX resistant to daraxonrasib and to zoldonrasib
- **Design:** CDK8 induces CXCL2 secretion and suppresses FAS expression, remodeling the microenvironment toward immune evasion after the initial T-cell influx that KRAS inhibition produces.
- **Effect size:** moderate — MRTX1133 first raised CD11c+ and T-cell infiltration near cancer cells, then that response reversed as CDK8-dependent resistance took hold. Targeting CDK8, alone or with anti-CTLA-4, overcame resistance, and CDK8 was also upregulated in daraxonrasib- and zoldonrasib-resistant PDX.
- **Variance:** vs vehicle and single-agent MRTX1133 arms; anti-CTLA-4 alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The primary models are G12D-driven; the daraxonrasib link rests on the PDX arm. No CDK8 inhibitor appears among the interventions surfaced for this case.

Mavridi/Joerger (2026) — Cell Death & Disease

Reference: [PMID 41748563](#) · Type: preclinical · Inclusion: included

- **Intervention:** rezatapopt (PC14586, p53 Y220C reactivator) — stability-response relationship
- **Indication / model:** recombinant p53 Y220C/Y220N/Y220S proteins with high-resolution crystal structures; matched mutant-expressing cell lines
- **Design:** Nanomolar binding into the mutation-induced crevice restores near-wild-type thermal stability, and how much stability comes back sets whether the cellular p53 program returns.
- **Effect size:** strong — Rezatapopt restored the folded conformation, target-gene transcription, antiproliferative

effect and apoptosis in Y220C cells, and in Y220S cells at higher concentrations. Y220N bound the drug with high-nanomolar affinity yet stayed too destabilized to respond, separating binding from functional rescue.

- **Variance:** vs wild-type p53 stability reference; untreated mutant lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Biophysics and cell lines only — no tumor model, no KRAS co-mutation arm, no PDAC context. Its value here is that Y220C specifically is the variant with full stability rescue.

Fece de la Cruz/Corcoran (2026) — Cancer Discovery

Reference: [PMID 41504628](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** rezatapopt (PC14586) — on-target acquired resistance
- **Indication / model:** functional modeling of engineered TP53 double mutants, anchored to patient ctDNA, biopsies and rapid-autopsy specimens at progression
- **Design:** Second TP53 alterations arising in cis with Y220C either abolish transcriptional activity outright or reshape the drug-binding surface so rezatapopt no longer engages.
- **Effect size:** moderate — Progression on rezatapopt was accompanied by heterogeneous secondary TP53 alterations in cis with Y220C. Engineered double mutants showed neither p53 reactivation nor target-gene induction on drug.
- **Variance:** vs Y220C single-mutant constructs; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Resistance is on-target and heterogeneous, so a second-generation Y220C reactivator would likely inherit much of it. The specimens are clinical; only the double-mutant modeling is preclinical.

Wang/Chen (2026) — Journal of Clinical Investigation

Reference: [PMID 42118591](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** KMT2C/COMPASS-complex loss (target-feature mechanism; no directed agent exists)
- **Indication / model:** conditional Kmt2c/Kmt2d knockout gastric GEMM, with and without Pten deletion; scRNA-seq integrated with human gastric cancer datasets
- **Design:** COMPASS-family loss reduces protein synthesis while raising ribosomal-protein transcription, leaving cells dependent on mTORC1; the same loss raises MHC class I and antigen presentation.
- **Effect size:** moderate — Kmt2c/d loss with Pten deletion produced muscle-invasive gastric adenocarcinoma within three weeks of Cre induction. Knockout cells were hypersensitive to mTORC1 inhibitors, and mTORC1 inhibition plus anti-PD-1 markedly suppressed tumors in immunocompetent mice.
- **Variance:** vs Kmt2c/d wild-type littermates; single-agent mTORC1 inhibition and anti-PD-1 arms; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Gastric lineage, double Kmt2c/d knockout, and Pten co-deletion — three departures from a single KMT2C S2816fs in MSS PDAC. No KMT2C-directed agent or basket trial exists, so this stays a hypothesis rather than a route for this patient.

Kyuno/Osanai (2026) — Tissue Barriers

Reference:PMID 40694660 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CLDN18.2 concordance across biopsy, resection and recurrent PDAC specimens (clone 43-14A)
- **Indication / model:** human PDAC tissue: 211 resections, 133 matched preoperative biopsies, 60 recurrent-lesion samples, all stained with the trial-validated 43-14A clone
- **Design:** Compares the same validated assay across specimen types to test whether an archival call transfers to a later or smaller sample.
- **n:** n=211 resections; n=133 matched biopsies; n=60 recurrent lesions
- **Effect size:** Biopsy-to-resection concordance was 92.5% at the 75% threshold, but sensitivity of biopsy for detecting positive tumors was only 54.6%, and lowering the biopsy threshold to 20% brought sensitivity to 100%. Recurrent lesions matched the primary in 83.3% of cases, with losses concentrated in local recurrence and liver metastasis.
- **Variance:** translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** The most decision-relevant row for the archival-versus-recurrence question here: expression usually persists at recurrence, but roughly one in six lesions diverges, and local recurrence is where loss clustered. Retrospective single-institution series.

Kerper/Graham (2026) — Virchows Archiv

Reference:PMID 42080920 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CLDN18.2 prevalence, primary-metastasis agreement, and neoadjuvant association in PDAC
- **Indication / model:** 337 resected human PDAC cases with paired metastases (102 pairs), scored by IHC against the gastroesophageal trial criteria
- **Design:** Prevalence and clinicopathologic correlation of CLDN18.2 staining, including relationship to prior neoadjuvant therapy.
- **n:** n=337 cases; n=102 primary-metastasis pairs
- **Effect size:** weak — 280 of 337 cases (83.1%) had some staining and 90 (26.7%) met the trial-grade criterion of moderate-to-strong staining in at least 75% of cells. Primary and metastatic tissue agreed in 76.5% of pairs. Positive cases were less often ones that had received neoadjuvant chemoradiation, and among neoadjuvant-treated patients positivity carried a modest survival advantage (HR 0.66).
- **Variance:** translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** The neoadjuvant association is cross-sectional and cannot separate treatment-induced antigen loss from selection of tumors that were negative to begin with. It is still the closest available signal on whether chemotherapy erodes the antigen, which matters after ten FOLFIRINOX cycles.

Al Subhi/Al-Mufargi (2026) — Frontiers in Medicine

Reference:PMID 42110438 · **Type:** preclinical · **Inclusion:** excluded — Aggregate of published IHC series rather than primary model or specimen data; its decision-relevant point, that positivity varies significantly by antibody clone, is already carried by the primary concordance studies retained here.

- **Intervention:** CLDN18.2 expression in PDAC — systematic review and meta-analysis
- **Toxicities:** n/a (preclinical)

- **Notes:** Excluded: Aggregate of published IHC series rather than primary model or specimen data; its decision-relevant point, that positivity varies significantly by antibody clone, is already carried by the primary concordance studies retained here.

Zhang/Ji (2026) — Advanced Science

Reference:PMID 41642951 · **Type:** preclinical · **Inclusion:** excluded — KMT2C-loss vulnerability reported in NF2-wild-type meningioma; neither the lineage nor the Hippo/ferroptosis axis has an established counterpart in PDAC, and no agent in the surfaced set acts on it.

- **Intervention:** KMT2C loss and ferroptosis sensitivity in meningioma
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: KMT2C-loss vulnerability reported in NF2-wild-type meningioma; neither the lineage nor the Hippo/ferroptosis axis has an established counterpart in PDAC, and no agent in the surfaced set acts on it.

Nishitani/Guo (2026) — bioRxiv (preprint)

Reference:PMID 42327198 · **Type:** preclinical · **Inclusion:** excluded — Un-peer-reviewed preprint in breast models. Its argument that MLL3's adaptor rather than catalytic role drives tumor suppression would predict catalytic-site inhibitors are futile, which is worth knowing but too preliminary and too far from PDAC to enter the synthesis.

- **Intervention:** MLL3 adaptor versus catalytic function in breast tumor suppression
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Un-peer-reviewed preprint in breast models. Its argument that MLL3's adaptor rather than catalytic role drives tumor suppression would predict catalytic-site inhibitors are futile, which is worth knowing but too preliminary and too far from PDAC to enter the synthesis.

Fu/Dumaz (2026) — Oncogene

Reference:PMID 42562950 · **Type:** preclinical · **Inclusion:** excluded — Melanoma models with no PDAC or KRAS G12R context, and RAF1 hotspot mutations do not appear on this patient's sequencing report.

- **Intervention:** RAF1 S257L/S259F and intrinsic daraxonrasib resistance in melanoma
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Melanoma models with no PDAC or KRAS G12R context, and RAF1 hotspot mutations do not appear on this patient's sequencing report.

Wang/Koong (2026) — bioRxiv (preprint)

Reference:PMID 42523398 · **Type:** preclinical · **Inclusion:** excluded — bioRxiv preprint pairing SBRT with KRAS inhibition; radiotherapy combination sits outside the modalities the trial screen surfaced for this case, and the work has not been peer reviewed.

- **Intervention:** Stereotactic radiotherapy combined with KRAS inhibition in PDAC
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: bioRxiv preprint pairing SBRT with KRAS inhibition; radiotherapy combination sits outside the modalities the trial screen surfaced for this case, and the work has not been peer reviewed.

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ASP546C (CLDN18.2 ADC)
- **Indication / model:** CLDN18.2-expressing gastroesophageal, pancreatic, and other solid tumors (2L (≥ 1 and ≤ 2 prior lines for advanced disease; pan-tumor cohort unrestricted)); Tumor expresses CLDN18.2; tissue required for central biomarker assessment
- **Design:** Phase 1/2 dose escalation and expansion of ASP546C
- **Effect size:** — — ORR; PK; safety
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ANO31905 (QLS31905, CLDN18.2 $\times$ CD3 bispecific) + GnP
- **Indication / model:** CLDN18.2-positive locally advanced unresectable or metastatic pancreatic cancer, first-line (1L); CLDN18.2-positive by central IHC (cutoff not stated in registry)
- **Design:** Phase 1b/2, ANO31905 (500-800 μ g/kg Q2W) + gemcitabine/nab-paclitaxel
- **Effect size:** — — DLT (1b); ORR (2)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** FG-M108 (anti-CLDN18.2 monoclonal antibody) + gemcitabine/nab-paclitaxel
- **Indication / model:** CLDN18.2-positive ($\geq 40\%$ of cells $\geq 2+$) unresectable/metastatic pancreatic cancer, first-line (1L); CLDN18.2 $\geq 40\%$ of tumor cells with membrane staining $\geq 2+$ by central IHC — the patient's 60% at 2+/3+ clears this
- **Design:** Randomized double-blind phase 3, FG-M108 + GnP vs placebo + GnP
- **Effect size:** — — OS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236) $\pm$ gemcitabine/nab-paclitaxel
- **Indication / model:** Metastatic pancreatic adenocarcinoma, first-line (RASolute 303) (1L (metastatic diagnosis ≤ 6 weeks before consent; no prior systemic therapy in the metastatic setting)); Documented RAS mutation

status, mutant or wild-type; G12R qualifies

- **Design:** Randomized phase 3, daraxonrasib monotherapy and daraxonrasib + GnP vs GnP
- **Effect size:** — — PFS/OS
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BLU-924 (SAR449336, pan-KRAS inhibitor)
- **Indication / model:** KRAS-mutant metastatic PDAC, NSCLC, or CRC (2L+); KRAS-mutant (allele coverage not itemized in registry); prior KRAS-targeted therapy excluded
- **Design:** First-in-human phase 1/2 dose escalation/enrichment/expansion of BLU-924
- **Effect size:** — — Safety/DLT; MTD
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** KST-6051 (pan-KRAS inhibitor)
- **Indication / model:** Advanced solid tumors with KRAS mutations, including PDAC, NSCLC, CRC (2L+); Documented KRAS mutation (no allele restriction stated) — G12R admissible as written
- **Design:** Phase 1 dose escalation of KST-6051 oral monotherapy
- **Effect size:** — — DLT; TEAEs
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** AN9025 (pan-RAS(ON) inhibitor)
- **Indication / model:** Advanced/metastatic solid tumors with KRAS/NRAS/HRAS mutations (2L+ (adjuvant therapy counts as a line if progression within 6 months of first dose)); KRAS/NRAS/HRAS mutation by validated tissue or cfDNA testing; prior pan-RAS(ON) exposure excluded
- **Design:** Phase 1 dose escalation of AN9025, an oral pan-RAS(ON) tri-complex inhibitor (same mechanism class as daraxonrasib)
- **Effect size:** — — DLT
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ELI-002 7P + tislelizumab + mFOLFIRINOX
- **Indication / model:** Resectable/borderline resectable localized PDAC with KRAS G12D/V/R/C/A/S or G13D (MSK, neoadjuvant) (neoadj); One of seven KRAS mutations including G12R; localized disease only, staging negative for metastasis
- **Design:** Phase 1, ELI-002 7P + tislelizumab + mFOLFIRINOX before surgery; ≤4 prior neoadjuvant mFOLFIRINOX doses allowed
- **Effect size:** — — Safety; immunologic response
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** zoldonrasib (RMC-9805, KRAS G12D(ON)-selective)
- **Indication / model:** Metastatic PDAC, first-line, KRAS G12D-mutant (1L); Documented KRAS G12D required — patient is G12R: excluded on allele
- **Design:** Randomized phase 3, zoldonrasib + investigator's-choice chemo vs placebo + chemo
- **Effect size:** — — PFS/OS
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TCR1188-ABC (KRAS G12V TCR-T)
- **Indication / model:** KRAS G12V-mutated pancreatic, cholangiocarcinoma, CRC, NSCLC (UPenn) (2L+); KRAS G12V plus HLA-A*11:01 required — patient is G12R: excluded on allele
- **Design:** Phase 1 of TCR1188-ABC base-edited autologous TCR-T with anti-ILT4 scFv armor, single infusion after Flu/Cy
- **Effect size:** — — Safety/DLT; MTD
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** LG00313112 (oral p53 Y220C reactivator)
- **Indication / model:** Advanced solid malignancies with TP53 Y220C mutation (2L+ (progression on most recent

line; refractory to or without standard options)); TP53 Y220C

- **Design:** Phase 1/2 dose escalation of LG00313112 once daily
- **Effect size:** — — DLT; safety
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** [68Ga]/[131I]-IR199 (CLDN18.2-directed iodine-131 radioconjugate, single dose)
- **Indication / model:** CLDN18.2-positive gastric and pancreatic cancer (n/a (biodistribution/dosimetry; ≥ 2 weeks from last myelosuppressive treatment)); CLDN18.2 expression required with ≥ 1 CLDN18.2-positive lesion; no numeric cutoff stated in the registry
- **Design:** Single-center first-in-human biodistribution and dosimetry study of a theranostic pair — [68Ga]-IR199 imaging plus a single dose of [131I]-IR199 therapeutic radioconjugate; estimated n=10
- **Effect size:** — — Absorbed radiation dose in tumors and normal organs (SPECT/CT dosimetry)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Wainberg/O'Reilly (2025) — Nat Med

Reference: [PMID 40790272](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** ELI-002 2P (lymph-node-targeted mKRAS G12D/G12R amphiphile vaccine)
- **Indication / model:** PDAC (n=20) and CRC (n=5) in minimal-residual-disease state after locoregional therapy (adj); MRD-positive (ctDNA/serum biomarker) after surgery and chemotherapy, radiographically recurrence-free. KRAS G12D or G12R tumors; the vaccine's peptide pool covers this patient's allele exactly.
- **Design:** Phase 1 single-arm (AMPLIFY-201, NCT04853017), final analysis at median 19.7 months follow-up
- **n:** 25
- **Effect size:** 0.12 HR — RFS by mKRAS T-cell response threshold
- **Variance:** P=0.0002 (threshold split, not a randomized comparison); p=0.0002
- **Toxicities:** dose-limiting toxicity (0/25, 0%)
- **Case fit:** weak
- **Notes:** Direct proof G12R-specific T-cell immunity can be raised in PDAC and that its magnitude tracks with relapse and survival. The threshold split is a biomarker correlation inside a single-arm trial, not a treatment-effect estimate. MRD setting does not match a radiographic recurrence; the class becomes relevant again after successful cytoreduction.

—/— (2025)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** zolbetuximab (anti-CLDN18.2 mAb) + gemcitabine/nab-paclitaxel
- **Indication / model:** CLDN18.2-positive ($\geq 75\%$ of tumor cells 2+/3+) metastatic PDAC, first-line (1L); 393 patients at 136 centers, all above the 75% CDx cutoff this patient's 60% fails.

- **Design:** Randomized phase 2 (GLEAM, NCT03816163), zolbetuximab + GnP vs GnP
- **n:** 393
- **Effect size:** 13.7 vs 13.6 months — OS — primary endpoint not met
- **Toxicities:** Sponsor states the combination's safety was consistent with the known profiles of each component; no per-term data released.
- **Case fit:** partial
- **Notes:** The most directly relevant CLDN18.2-antibody efficacy datum in PDAC, and it is negative: no OS gain even in $\geq 75\%$ expressers. Source is the Astellas topline release of 2025-10; no peer-reviewed publication found and the CT.gov record has no posted results (both checked August 2026), so per-term AE extraction was not possible. Zolbetuximab here fails on eligibility ($60\% < 75\%$) and on efficacy; the CLDN18.2 strategy routes through CAR-T, engagers, and ADCs instead.

Qi/Shen (2025) — Lancet

Reference: [PMID 40460847](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** satri-cel (CT041, autologous CLDN18.2 CAR-T)
- **Indication / model:** CLDN18.2-positive ($2+$ intensity in $\geq 40\%$ of cells) advanced gastric/GEJ adenocarcinoma, ≥ 2 prior lines (2L+); 156 randomized 2:1 in China; 69% of the satri-cel group had peritoneal metastasis. The $\geq 2+$ in $\geq 40\%$ eligibility rule is the one this patient's 60% at 2+/3+ clears.
- **Design:** Randomized open-label phase 2 (CT041-ST-01, NCT04581473)
- **n:** 156
- **Effect size:** 3.25 months — PFS (IRC, ITT)
- **Variance:** 95% CI 2.86–4.53; $p < 0.0001$
- **Toxicities:** any TEAE $G \geq 3$ (87/88, 99%); lymphocyte count decreased $G \geq 3$ (86/88, 98%); white blood cell count decreased $G \geq 3$ (68/88, 77%); neutrophil count decreased $G \geq 3$ (58/88, 66%); CRS (84/88, 95%)
- **Case fit:** cross_tumor_only
- **Notes:** First randomized CAR-T win in a solid tumor; drove the June 2026 NMPA approval in 2L+ gastric cancer. Anchors the CLDN18.2 CAR-T strategy mechanistically, but the tumor type differs and the modest absolute PFS should temper extrapolation to PDAC.

—/— (2025) — Ann Oncol (ESMO 2025 abstract 2137P)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** ASP2138 (CLDN18.2xCD3 2+1 bispecific T-cell engager)
- **Indication / model:** CLDN18.2-positive advanced gastric/GEJ adenocarcinoma and metastatic PDAC (phase 1/1b monotherapy) (2L+); Previously treated G/GEJ and PDAC; monotherapy escalation cutoff $\geq 1+$ membranous staining, far below this patient's 2+/3+ in 60%.
- **Design:** Phase 1/1b dose escalation/expansion, interim analysis (NCT05365581); ESMO 2025 abstract 2137P
- **Effect size:** 41.7 % — 12-week unconfirmed disease-control rate, G/GEJ ≥ 420 ug
- **Toxicities:** any TRAE (77.3%); CRS (43.2%); nausea (44.7%); vomiting (29.8%); pyrexia (23.4%)
- **Case fit:** partial
- **Notes:** Conference abstract only; the PDAC cohort has enrolled and contributes safety data, but the efficacy readout shown so far is gastric. Congress-abstract DOI could not be verified against Crossref, so identifiers stay null; source is Annals of Oncology 2025 supplement, abstract 2137P (publisher page blocked robots at review).

Liu/Shen (2025) — Nat Med

Reference: [PMID 40670773](#) · Type: clinical · Inclusion: included

- **Intervention:** IBI343 (CLDN18.2 ADC, exatecan payload)
- **Indication / model:** Advanced gastric/GEJ adenocarcinoma, CLDN18.2-positive (phase 1) (2L+); 127 dosed (19 escalation, 108 expansion); headline efficacy in CLDN18.2-high (2+/3+ $\geq 75\%$) at 6 mg/kg.
- **Design:** Phase 1 dose escalation/expansion, single-arm (NCT05458219)
- **n:** 127
- **Effect size:** 29 % — confirmed ORR, CLDN18.2-high 6 mg/kg
- **Variance:** 95% CI 14.2–48.0
- **Toxicities:** any TRAE $G \geq 3$ (n=116, 52.6%); neutrophil count decreased $G \geq 3$ (n=116, 28.4%); white blood cell count decreased $G \geq 3$ (n=116, 25.9%); anemia $G \geq 3$ (n=116, 16.4%); ILD/pneumonitis (0/127, 0%); treatment-related death $G5$ (0/116, 0%)
- **Case fit:** cross_tumor_only
- **Notes:** Gastric phase 1 that set the 6 mg/kg dose now in phase 3 in both gastric and pancreatic cancer. AE data from PMC full text (PMC12443601).

—/— (2025) — J Clin Oncol (suppl; ASCO 2025 abstract 4017)

Reference: [doi:10.1200/JCO.2025.43.16_suppl.4017](#) · Type: clinical · Inclusion: included

- **Intervention:** IBI343 (CLDN18.2 ADC, exatecan payload)
- **Indication / model:** Previously treated advanced PDAC, CLDN18.2-expressing (phase 1 dose-expansion) (2L+); 83 PDAC patients dosed overall; efficacy cohort is 44 at 6 mg/kg with CLDN18.2 (1+/2+/3+) $\geq 60\%$ — the boundary this patient's 60% sits exactly on, though the tier composition differs from the 2+/3+ call on the resection.
- **Design:** Phase 1 dose-expansion cohort, single-arm (NCT05458219); ASCO 2025 oral, abstract 4017
- **n:** 83
- **Effect size:** 22.7 % — confirmed ORR, 6 mg/kg, CLDN18.2 $\geq 60\%$
- **Toxicities:** any TEAE (n=83, 98.8%); anemia (n=83); neutrophil count decreased (n=83); nausea/vomiting $G \geq 3$ (0/83, 0%)
- **Case fit:** strong
- **Notes:** The largest prospective CLDN18.2-directed efficacy dataset in previously treated PDAC to date, and the direct basis for the randomized phase 3 this patient could screen for (NCT07066098). Abstract plus sponsor release; per-term grade ≥ 3 rates were not published, and the abstract page returned 403 at review.

Ruan/Xu (2025) — Lancet Oncol

Reference: [PMID 39788133](#) · Type: clinical · Inclusion: included

- **Intervention:** AZD0901 / CMG901 (CLDN18.2 ADC, MMAE payload)
- **Indication / model:** Advanced gastric/GEJ adenocarcinoma, CLDN18.2-positive (first-in-human phase 1) (2L+); 27 escalation + 107 gastric/GEJ expansion in China; refractory to or lacking standard therapy; pancreatic dose-expansion data held for a separate publication.
- **Design:** Phase 1 dose escalation/expansion, single-arm (KYM901, NCT04805307)
- **n:** 134
- **Effect size:** 28 % — confirmed ORR, 2.2-3.0 mg/kg G/GEJ full analysis set

- **Variance:** 95% CI 20–38
- **Toxicities:** any TEAE G $\geq$ 3 (73/107, 68%); neutrophil count decreased G3 (22/107, 21%); anemia G3 (15/107, 14%); vomiting G3 (11/107, 10%); pancreatitis (DLT) G3 (1/27); treatment-related death G5 (1/107)
- **Case fit:** cross_tumor_only
- **Notes:** First-in-human basis for AstraZeneca's AZD0901 program, whose multi-cohort phase 2 (NCT06219941) names PDAC. The pancreatic expansion cohort exists but its data are unpublished — a gap worth re-checking.

Ku/Klempner (2025) — Clin Cancer Res

Reference: [PMID 40586719](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** givastomig (TJ033721/ABL111, CLDN18.2x4-1BB bispecific)
- **Indication / model:** CLDN18.2-positive advanced solid tumors; efficacy population gastroesophageal cancer (2L+); 75 enrolled; CLDN18.2-positive defined as $\geq$ 1+ on $\geq$ 1% of cells — responders spanned 11% to 100% expression, i.e. activity well below this patient's 60%.
- **Design:** First-in-human phase 1 dose escalation/expansion (NCT04900818)
- **n:** 75
- **Effect size:** 16 % — ORR, CLDN18.2+ GEC at $\geq$ 5 mg/kg
- **Toxicities:** dose-limiting toxicity (0/36, 0%); nausea (n=75); anemia (n=75); fatigue (n=75)
- **Case fit:** cross_tumor_only
- **Notes:** Mechanism evidence that a 4-1BB-conditional CLDN18.2 engager is active across a wide expression range without CD3-engager toxicity. Development is gastroesophageal-centered; no visible PDAC route today.

—/— (2025) — J Clin Oncol (suppl; ASCO GI 2025 abstract 729)

Reference: [doi:10.1200/JCO.2025.43.4_suppl.729](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** FG-M108 (ADCC-enhanced anti-CLDN18.2 mAb) + gemcitabine/nab-paclitaxel
- **Indication / model:** CLDN18.2-positive (IHC 1+/2+/3+ $\geq$ 10%) untreated locally advanced/metastatic pancreatic cancer (1L); 50 patients (39 in C2, 11 in D2), data cut September 5, 2024.
- **Design:** Phase 1b single-arm dose cohorts (NCT04894825); ASCO GI 2025 abstract 729
- **n:** 50
- **Effect size:** 32.4 % — ORR, cohort C2
- **Toxicities:** anemia (n=39, 56.2%); nausea (n=39, 56.4%); vomiting (n=39, 48.7%); hypoalbuminemia (n=39, 46.2%)
- **Case fit:** partial
- **Notes:** Phase 1b behind the first PDAC phase 3 whose cutoff ($\geq$ 40% at $\geq$ 2+, NCT07383922) this tumor actually clears. Abstract-level data; single-arm ORR on a chemo backbone cannot isolate the antibody's contribution.

—/— (2025)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** JAB-30355 (oral p53 Y220C reactivator)
- **Indication / model:** Advanced solid tumors with TP53 Y220C (phase 1/2a, NCT06386146) (2L+); Dose escalation in the US and China; first patient dosed July 2024; registry criteria carry no KRAS co-mutation exclusion.

- **Design:** Phase 1/2a dose escalation/expansion, ongoing; no results published
- **Effect size:** — — sponsor-reported efficacy signal; no data disclosed
- **Case fit:** strong
- **Notes:** The practical Y220C route for this KRAS-co-mutated patient, and its entire efficacy record is one sponsor phrase: 'positive efficacy signal' at high dose in the August 2025 interim report, with the phase 1 readout guided to H1 2026 and nothing published as of August 2026. No paper, no abstract, no CT.gov results — toxicity and effect-size fields are empty because no source exists, not because extraction was skipped. — (toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** clinical · **Inclusion:** excluded — program discontinued (Elevation Oncology halted EO-3021 in 2025 after gastric expansion ORR of 22.2% underwhelmed); no PDAC development path

- **Intervention:** EO-3021 / SYSA1801 (CLDN18.2 ADC, MMAE payload)
- **Notes:** Excluded: program discontinued (Elevation Oncology halted EO-3021 in 2025 after gastric expansion ORR of 22.2% underwhelmed); no PDAC development path — Reviewed at press-release/abstract level only; kept on the ledger as expectation-setting for the ADC class at moderate expression.

—/— (2025)

Reference: — · **Type:** clinical · **Inclusion:** excluded — eligibility itemizes KRAS G12A/C/D/S/V and omits G12R despite the pan-KRAS label; no peer-reviewed clinical results published

- **Intervention:** BBO-11818 (pan-KRAS inhibitor)
- **Notes:** Excluded: eligibility itemizes KRAS G12A/C/D/S/V and omits G12R despite the pan-KRAS label; no peer-reviewed clinical results published — The cautionary example that 'pan-KRAS' labels need codon-level checking; documented in trials.jsonl (NCT06917079).

—/— (2025)

Reference: — · **Type:** clinical · **Inclusion:** excluded — no published clinical evidence — first-in-human dose escalation (NCT07094204) opened 2025; G12R is explicitly eligible, so this is a trial route, not an evidence base

- **Intervention:** ASP5834 (IV pan-KRAS inhibitor)
- **Notes:** Excluded: no published clinical evidence — first-in-human dose escalation (NCT07094204) opened 2025; G12R is explicitly eligible, so this is a trial route, not an evidence base — PubMed and congress searches returned preclinical material only, which the researcher track owns.

—/— (2025)

Reference: — · **Type:** clinical · **Inclusion:** excluded — no published clinical data of any kind — first-in-human NCT07060989 opened 2025; no abstract, publication, or sponsor efficacy statement found

- **Intervention:** NTS071 (oral p53 Y220C reactivator)
- **Notes:** Excluded: no published clinical data of any kind — first-in-human NCT07060989 opened 2025; no abstract, publication, or sponsor efficacy statement found — Second US-recruiting Y220C option without a registry-level KRAS exclusion; evidence base currently empty, thinner even than JAB-30355's

Puzio-Kuter/Poyurovsky (2025) — Cancer Discovery

Reference:PMID 39945593 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** rezatapopt (PC14586, p53 Y220C reactivator)
- **Indication / model:** Y220C-mutant human tumor cell lines and xenografts; biochemical and structural analysis; in vitro and in vivo transcriptomics
- **Design:** The compound occupies the crevice the Y220C substitution creates, correcting the fold of the DNA-binding domain so p53 binds DNA and drives its wild-type target program again.
- **Effect size:** strong — Multiple arms of the wild-type p53 program were recovered in cells and in tumors, with antiproliferative effect. Antitumor activity held as monotherapy and improved in combination with immunotherapy.
- **Variance:** vs vehicle; p53 wild-type and non-Y220C mutant lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** No PDAC model and no KRAS-mutant background is reported, which is precisely the gap the clinical KRAS-wild-type responder pattern points at.

Xu/Bader (2025) — Cancer Immunology Research

Reference:PMID 39404622 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** TAC01-CLDN18.2 (T-cell antigen coupler engineered T cells)
- **Indication / model:** CLDN18.2-expressing gastric, gastroesophageal and pancreatic 2D cultures and spheroids; xenograft mice; primary human cells from vital organs
- **Design:** The T-cell antigen coupler routes antigen recognition through the endogenous TCR complex rather than a synthetic signalling chain, which removes tonic signalling.
- **Effect size:** strong — Cytotoxicity tracked with CLDN18.2 level yet was still present in low-antigen models. Established xenografts were eradicated with durable responses on recursive killing and rechallenge, and no reactivity appeared against normal-tissue cells.
- **Variance:** vs non-transduced T cells; CLDN18.2-negative lines; normal-tissue cocultures; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Sponsor preclinical package supporting the phase 1/2. Xenografts in immunodeficient mice leave stromal and host-immune barriers untested, and the clean normal-cell result sits uneasily beside the gastric toxicity seen with CAR constructs.

Zhang/Yu (2025) — Oncogene

Reference:PMID 40301544 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CLDN18.2 STAR-T versus CAR-T in pancreatic cancer
- **Indication / model:** pancreatic cancer models in vivo comparing CLDN18.2 STAR-T with CLDN18.2 CAR-T; gastroscopic assessment in treated patients
- **Design:** Synthetic TCR-and-antigen receptor signalling through the endogenous CD3 complex gives lower-gain activation than a second-generation CAR, trading potency for less normal-gastric damage; IL-12 p40 armoring recovers part of the potency.

- **Effect size:** moderate — In vitro cytotoxicity matched CAR-T, but in vivo STAR-T caused less gastric damage and had weaker antitumor effect. Adding the IL-12beta p40 subunit improved STAR-T function in vitro and in mice.
- **Variance:** vs CLDN18.2 CAR-T; untransduced T cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** The explicit potency-versus-gastric-toxicity trade is the honest framing for CLDN18.2 cell therapy in PDAC. The accompanying patient experience is small and uncontrolled.

Birocchi/Maus (2025) — Nature Communications

Reference: [PMID 41176533](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CLDN18.2 CAR-T on-target off-tumor gastric toxicity model
- **Indication / model:** immunocompetent mouse model built on CLDN18.2 epitopes shared between human and mouse, dosed with CLDN18.2 CAR-T of varying design
- **Design:** CAR-T recognize CLDN18.2 on normal gastric mucosa, and the resulting injury does not depend on construct design, costimulatory domain, or tumor model.
- **Effect size:** moderate — The model reproduced gastric on-target off-tumor toxicity across constructs and served as a testbed for Boolean AND-gate mitigation. The authors caution that such models can overcall severity relative to patients as easily as undercall it.
- **Variance:** vs untransduced T cells; CAR variants differing in costimulatory domain; AND-gate logic constructs; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Gastric toxicity is the practical dose ceiling for every CLDN18.2 cell therapy in this file. How it behaves in altered post-resection upper-GI anatomy is not modelled here.

Gaspar/Cobbold (2025) — Journal for ImmunoTherapy of Cancer

Reference: [PMID 40759445](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** AZD5863 (CLDN18.2xCD3 affinity-modulated T-cell engager)
- **Indication / model:** CLDN18.2-expressing human lines cocultured with human or cynomolgus PBMC; humanized and human-CD3 transgenic mice bearing pancreatic, gastric and esophageal tumors
- **Design:** High CLDN18.2 affinity paired with deliberately low CD3 affinity preserves killing while cutting peripheral cytokine release; killing spreads to antigen-negative neighbours partly through IFN-gamma, TNF-alpha and Fas ligand.
- **Effect size:** strong — Potency correlated significantly with CLDN18.2 receptor density. Bystander killing of CLDN18.2-negative cells occurred whenever positive cells were present, and tumor control was achieved in pancreatic as well as gastric and esophageal models.
- **Variance:** vs a CLDN18.2 engager with higher CD3 affinity; CLDN18.2-negative lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Density dependence cuts both ways for a tumor staining 2+/3+ in only 60% of cells — the bystander arm is why heterogeneity may still be tractable, but this is cell-line xenograft work without stroma-rich or organoid modelling.

Carstens/Smith (2025) — Nature Communications

Reference:PMID 41176582 · **Type:** preclinical · **Inclusion:** excluded — Reaches the same conclusion as Birocchi (PMID 41176533) about gastric on-target injury from CLDN18.2-directed therapy; the Birocchi row was kept because it also tests AND-gate mitigation, and logging two overlapping models would double-count one mechanism.

- **Intervention:** CLDN18.2 on-target/off-tumor toxicity modeling (Carstens)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Reaches the same conclusion as Birocchi (PMID 41176533) about gastric on-target injury from CLDN18.2-directed therapy; the Birocchi row was kept because it also tests AND-gate mitigation, and logging two overlapping models would double-count one mechanism.

Ma/Qi (2025) — iScience

Reference:PMID 39925434 · **Type:** preclinical · **Inclusion:** excluded — Correlative single-cell analysis of blood and ascites from five patients already on a CAR-T trial, not a preclinical model; the naive-like-T-cell and CLDN18-high/MYC-high prognostic splits are hypothesis-generating at n=5.

- **Intervention:** satri-cel (CT041) response correlates by single-cell sequencing
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Correlative single-cell analysis of blood and ascites from five patients already on a CAR-T trial, not a preclinical model; the naive-like-T-cell and CLDN18-high/MYC-high prognostic splits are hypothesis-generating at n=5.

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IBI343 (CLDN18.2 ADC)
- **Indication / model:** Previously treated, CLDN18.2+ pancreatic cancer (2L+); CLDN18.2-positive per sponsor assay (cutoff not stated in the registry record)
- **Design:** Randomized double-blind phase 3, IBI343 6 mg/kg Q3W vs placebo
- **Effect size:** — — OS
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IBI343 + gemcitabine + nab-paclitaxel
- **Indication / model:** CLDN18.2+ advanced pancreatic cancer, first-line (1L); CLDN18.2-positive by pathological testing
- **Design:** Phase 2 single-arm, IBI343 + gemcitabine/nab-paclitaxel
- **Effect size:** — — ORR; safety
- **Case fit:** partial

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** AZD4360 (CLDN18.2 ADC)
- **Indication / model:** CLDN18.2+ advanced PDAC, gastric/GEJ, and biliary tract cancer (2L+ (≥1 prior line for advanced/metastatic disease)); Documented positive CLDN18.2 expression
- **Design:** Phase 1/2 dose escalation and expansion of AZD4360
- **Effect size:** — — Safety/DLT
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** QLS31905 (CLDN18.2×CD3 bispecific) + gemcitabine/nab-paclitaxel
- **Indication / model:** CLDN18.2-positive pancreatic cancer, first-line (China) (1L (no prior systemic therapy for unresectable/metastatic disease)); CLDN18.2-positive (sponsor assay)
- **Design:** Randomized double-blind phase 3, QLS31905 + GnP vs placebo + GnP
- **Effect size:** — — OS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ASP5834 (pan-KRAS inhibitor, IV)
- **Indication / model:** KRAS-mutant solid tumors with PDAC expansion cohorts (2L (PDAC expansion: ≤2 prior advanced-setting lines; neoadjuvant/adjuvant with recurrence ≤6 months counts as one line)); KRAS G12V, G12D, G12C, G12R, G12A, or G13D by local testing — G12R is named explicitly
- **Design:** Phase 1 dose escalation/expansion of ASP5834 (IV) ± panitumumab
- **Effect size:** — — Safety/DLT
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** JAB-23E73 (oral pan-KRAS inhibitor)

- **Indication / model:** Advanced solid tumors with KRAS alteration (2L+); KRAS alteration (pan-KRAS; no allele list in registry); prior KRAS/pan-KRAS inhibitor exposure excluded
- **Design:** Phase 1/2a dose escalation and expansion of JAB-23E73 monotherapy (US/global protocol; a China counterpart runs as NCT06959615 and a PDAC chemo-combination study as NCT07640295)
- **Effect size:** — — MTD/RP2D; safety
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** JYP0015 (ERAS-0015, pan-RAS(ON) molecular glue)
- **Indication / model:** Advanced solid tumors with RAS mutations, PDAC named (2L+); RAS-mutant (KRAS/NRAS/HRAS, active-state ON inhibitor) — G12R within scope
- **Design:** Phase 2 of JYP0015 (ERAS-0015) oral pan-RAS molecular glue
- **Effect size:** — — DLT; ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BBO-11818 (pan-KRAS inhibitor)
- **Indication / model:** KRAS-mutant NSCLC, PDAC, CRC and other solid tumors (2L+); Eligible alleles G12A, G12C, G12D, G12S, G12V — G12R is absent from the list: excluded on allele
- **Design:** Phase 1 of BBO-11818 pan-KRAS inhibitor ± pembrolizumab, chemo, cetuximab, or BBO-10203
- **Effect size:** — — Safety/DLT
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ANOC-001 / ANOC-002 / ANOC-003 (mutant-KRAS TCR-T)
- **Indication / model:** Metastatic/locally advanced PDAC (Anocca TCR-T master protocol) (2L+); KRAS G12V (ANOC-001/-002) or G12D (ANOC-003) with matching HLA alleles — G12R not covered: excluded on allele
- **Design:** Master protocol, gene-edited autologous TCR-T targeting HLA-presented mutant-KRAS peptides, single infusion after Flu/Cy
- **Effect size:** — — DLT; MTD/RP2D
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NTS071 (oral p53 Y220C reactivator)
- **Indication / model:** Advanced solid tumors with TP53 Y220C mutation (2L+ (≥1 prior line with progressive disease)); TP53 Y220C; no KRAS co-mutation exclusion in registry
- **Design:** Phase 1/2a dose escalation and expansion of NTS071 monotherapy
- **Effect size:** — — DLT; ORR (phase 2a)
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** GenSci128 (p53 Y220C reactivator)
- **Indication / model:** Solid tumors harboring a TP53 Y220C mutation (China) (2L+); Confirmed TP53 Y220C in tumor tissue; prior Y220C-reactivator exposure excluded
- **Design:** First-in-human phase 1 of GenSci128 tablets
- **Effect size:** — — DLT; MTD/RP2D
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TNhYP218 (mesothelin CAR-T)
- **Indication / model:** Mesothelin-expressing solid tumors including pancreatic cancer (NCI) (2L+ (progression on ≥1 approved systemic therapy; no line limit)); Mesothelin 2+/3+ in ≥50% of cells by IHC on fresh or archival tissue — not yet measured in this patient
- **Design:** Phase 1 of TNhYP218 (naive/stem-cell-memory-enriched anti-mesothelin hYP218 CAR-T) after Flu/Cy
- **Effect size:** — — RP2D; ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** 177Lu-DOTA-SNA040 (CLDN18.2-directed lutetium-177 radioligand therapy)
- **Indication / model:** CLDN18.2-positive advanced solid tumors — gastric/GEJ, pancreatic cancer, cholangiocarcinoma — after first-line failure or progression (2L+ (first-line treatment failure or progression)); CLDN18.2 IHC membrane staining ≥2+ in ≥20% of tumor cells in at least one lesion — the patient's 2+/3+ in

60% clears this cutoff three times over

- **Design:** Single-center open-label first-in-human dose-escalation IIT of a single dose of ¹⁷⁷Lu-labeled CLDN18.2-directed radioligand; estimated n=12
- **Effect size:** — — DLT; safety and tolerability; biodistribution
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Qi/Shen (2024) — Nat Med

Reference:PMID 38830992 · **Type:** clinical · **Inclusion:** included

- **Intervention:** satri-cel (CT041, autologous CLDN18.2 CAR-T)
- **Indication / model:** CLDN18.2-positive advanced gastrointestinal cancers (gastric-dominant; includes pancreatic patients) (2L+); 98 infused across escalation and four expansion cohorts (monotherapy, +anti-PD-1, post-1L sequential, anti-CLDN18.2-mAb-refractory).
- **Design:** Phase 1 single-arm, final results (CT041-CG4006, NCT03874897); median follow-up 32.4 months from apheresis
- **n:** 98
- **Effect size:** 38.8 % — ORR
- **Toxicities:** CRS (n=98, 96.9%); gastric mucosal injury (8/98, 8.2%); ICANS (0/98, 0%); treatment-related death G5 (0/98, 0%)
- **Case fit:** partial
- **Notes:** The long-follow-up single-arm base for satri-cel across GI cancers; the two published PDAC responders (PMID 37689733) were treated under this program's protocols. Supersedes the 2022 interim report.

Pant/O'Reilly (2024) — Nat Med

Reference:PMID 38195752 · **Type:** clinical · **Inclusion:** excluded — superseded by the final analysis of the same trial (AMPLIFY-201, PMID 40790272)

- **Intervention:** ELI-002 2P (mKRAS amphiphile vaccine)
- **Notes:** Excluded: superseded by the final analysis of the same trial (AMPLIFY-201, PMID 40790272) — Interim publication; retained as the safety source (no DLTs, RP2D 10 mg Amph-CpG-7909) cited in the final-results row.

Holderfield/Singh (2024) — Nature

Reference:PMID 38589574 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236) / RMC-7977 tool analogue, pan-RAS(ON) multi-selective inhibitor
- **Indication / model:** RAS-addicted human cancer cell lines and xenografts across RAS genotypes, weighted to KRAS codon-12 mutants
- **Design:** RMC-7977 assembles a tri-complex with cyclophilin A and the GTP-bound (ON) state of KRAS, NRAS and HRAS, sterically blocking effector engagement by mutant and wild-type RAS simultaneously.
- **Effect size:** strong — Tumor regression at tolerated exposures across RAS-addicted models, deepest in KRAS codon-12 mutants. Models that had escaped G12C(OFF) inhibitors by restoring RAS pathway signalling remained sensitive.
- **Variance:** vs vehicle; KRAS G12C(OFF) inhibitor comparison in G12C-resistant models; translatability: high

- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** RMC-7977 is the tool analogue of the clinical agent. Activity is reported for KRAS G12X in aggregate with no G12R-specific arm, which is the gap the whole clinical estimand for this patient inherits.

Wasko/Olive (2024) — Nature

Reference: [PMID 38588697](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236) / RMC-7977 tool analogue, pan-RAS(ON) multi-selective inhibitor
- **Indication / model:** autochthonous KPC mice, PDAC organoids and cell lines, xenografts
- **Design:** Broad RAS-GTP inhibition collapses tumor RAS output while normal epithelium tolerates it, so tumors take waves of apoptosis plus sustained proliferative arrest and normal tissue only slows transiently.
- **Effect size:** strong — Anti-tumor activity was broad across PDAC models at well-tolerated exposures. KPC survival was extended substantially before on-treatment relapse, and relapsed tumors were enriched for Myc copy gain, which TEAD inhibition countered in vitro.
- **Variance:** vs vehicle-treated KPC and xenograft cohorts; normal tissue used as the selectivity comparator; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** KPC is Kras G12D-driven, so allele transfer to G12R is an assumption rather than a result. An author correction was issued in 2024 (PMID 39533066) without altering the conclusions.

Drakes/DeMuth (2024) — Cancer Immunology Research

Reference: [PMID 38270373](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ELI-002 amphiphile vaccine platform (lymph-node-targeted mKRAS peptide + CpG)
- **Indication / model:** murine syngeneic solid-tumor models with adoptive TCR-T transfer; human TCR-T in vitro against NY-ESO-1, mutant KRAS and HPV16 E7
- **Design:** Amphiphile conjugation traffics peptide and CpG adjuvant into draining lymph nodes, maturing antigen-presenting cells and expanding both transferred and endogenous antigen-specific T cells without lymphodepletion.
- **Effect size:** strong — Vaccine plus TCR-T eradicated established tumors that resisted TCR-T alone, with durable protection and antigen spreading to antigens outside the vaccine. Matched amphiphile peptides improved proliferation, activation and cytotoxicity of human mutant-KRAS TCR-T in vitro.
- **Variance:** vs TCR-T monotherapy; unvaccinated controls; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Not a PDAC model and no G12R-specific in-vivo arm, though the clinical peptide pool covers that allele. Platforms of this class were built for low tumor burden, which a growing radiographic recurrence is not.

Barrett/Bosco (2024) — Clinical Cancer Research

Reference: [PMID 39321207](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** AZD6422 (TGF-beta-armored CLDN18.2 CAR-T) — class mechanism for stromal suppression
- **Indication / model:** patient-derived xenografts of gastric, esophageal and pancreatic tumors spanning a range

of CLDN18.2 and TGF-beta levels

- **Design:** A CD28/CD3z CAR against CLDN18.2 carrying a dominant-negative TGF-beta receptor, so stromal TGF-beta cannot suppress the transferred cells; a shortened manufacturing process preserves potency.
- **Effect size:** strong — Antitumor activity held across PDX with varying antigen and TGF-beta levels, and the armoring advantage widened in high-TGF-beta tumors. Restimulation assays showed better persistence and cytolytic function than a conventionally manufactured product.
- **Variance:** vs conventionally manufactured CLDN18.2 CAR-T without TGF-beta armoring; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Immunodeficient PDX cannot report on the desmoplastic barrier or host immunity; TGF-beta armoring addresses one component of that barrier. AZD6422 itself is not among the trials screened for this case, so this is class evidence.

Xu/Chen (2024) — Cell Reports Medicine

Reference: [PMID 39232496](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** AZD0901 / CMG901 (CLDN18.2 ADC, MMAE payload)
- **Indication / model:** CLDN18.2-expressing tumor cell lines and cocultures; patient-derived xenografts; rat and cynomolgus toxicology
- **Design:** MMAE delivered on a CLDN18.2 antibody kills through direct cytotoxicity plus ADCC, CDC, and bystander killing of neighbouring cells.
- **Effect size:** strong — Binding was CLDN18.2-specific and killing ran through all four mechanisms in vitro, with significant antitumor activity in PDX. Toxicology was dominated by reversible MMAE-type hematopoietic effects.
- **Variance:** vs isotype ADC; CLDN18.2-negative cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** PDX work is gastric-weighted with no separately reported pancreatic arm and no expression-threshold analysis, so the antigen level needed for activity is not pinned down here.

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236) + chemotherapy
- **Indication / model:** RAS-mutated gastrointestinal tumors including PDAC; daraxonrasib + chemotherapy combinations (1L/2L+ (subprotocol-specific)); RAS-mutant; the G12D restriction applies only to subprotocols D/E/F, not the daraxonrasib chemo-combination arms
- **Design:** Phase 1/2 master protocol, RAS(ON) inhibitors + mFOLFIRINOX, mFOLFOX6, GnP, bevacizumab, or cetuximab
- **Effect size:** — — Safety/DLT
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** PF-07934040 (pan-KRAS inhibitor)
- **Indication / model:** KRAS-mutant PDAC (any variant), CRC, NSCLC; monotherapy and combinations including GnP (1L-3L (cohort-specific; PDAC 2-3L cohort counts neoadjuvant/adjuvant per protocol rules)); Confirmed KRAS mutation, any variant, for PDAC — G12R covered
- **Design:** Phase 1 dose escalation/expansion of PF-07934040 alone and with chemotherapy or cetuximab
- **Effect size:** — — Safety/DLT
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** mKRASvax + balstilimab + botensilimab
- **Indication / model:** Stage IV MMR-proficient CRC and PDAC with vaccine-covered KRAS mutations; maintenance and KRAS-inhibitor-exposed cohorts (2L+/maintenance); Tumor expresses one of the pooled vaccine's KRAS mutations (G12R included); MMR-proficient — the patient's MSS status is required here, not exclusionary
- **Design:** Phase 1/2, pooled mutant-KRAS long-peptide vaccine with poly-ICLC + balstilimab (anti-PD-1) + botensilimab (Fc-enhanced anti-CTLA-4)
- **Effect size:** — — PFS (mPDAC maintenance cohort); ORR (other cohorts)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** daratumumab + TG01 + nivolumab
- **Indication / model:** PDAC after one failed treatment (and NSCLC post-anti-PD-1) with KRAS codon 12/13 mutations (2L); KRAS codon 12 (12A/C/D/R/S/V) or 13D — G12R named
- **Design:** Phase 2, daratumumab (anti-CD38) + TG01 mKRAS vaccine + nivolumab
- **Effect size:** — — ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NT-112 / AZD0240 (KRAS G12D-directed TCR-T)
- **Indication / model:** Unresectable/metastatic solid tumors incl. PDAC with KRAS G12D (2L+); KRAS G12D plus HLA-C08:02 or HLA-A11:01/02 required — patient is G12R (HLA untyped): excluded on allele
- **Design:** Phase 1 of NT-112 and AZD0240 autologous KRAS G12D TCR-T cells
- **Effect size:** — — Safety; MTD/RDE

- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** JAB-30355 (oral p53 Y220C reactivator)
- **Indication / model:** Advanced solid tumors with TP53 Y220C mutation (2L+ (≥1 prior line of systemic therapy for tumor type and stage)); Documented TP53 Y220C; no KRAS co-mutation exclusion appears in the registry record
- **Design:** Phase 1/2a dose escalation and expansion of JAB-30355 monotherapy
- **Effect size:** — — DLT; safety
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** sacituzumab tirumotecan (sac-TMT, MK-2870/SKB264, TROP2 ADC)
- **Indication / model:** Previously treated advanced PDAC, CRC, and biliary tract cancer (MK-9999 GI umbrella) (2L+); No TROP2 expression gate; TROP2 is broadly expressed in PDAC but unmeasured in this patient and not a nominated feature
- **Design:** Phase 1/2 umbrella, sacituzumab tirumotecan alone or in combinations, PDAC cohort included
- **Effect size:** — — Safety/DLT; ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Shitara/Ajani (2023) — Lancet

Reference: [PMID 37068504](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** zolbetuximab (anti-CLDN18.2 mAb) + mFOLFOX6
- **Indication / model:** CLDN18.2-positive (≥75% 2+/3+), HER2-negative, untreated advanced gastric/GEJ adenocarcinoma (1L); 565 randomized across 215 centers; ECOG 0-1.
- **Design:** Randomized double-blind placebo-controlled phase 3 (SPOTLIGHT, NCT03504397)
- **n:** 565
- **Effect size:** 10.61 months — PFS (IRC)
- **Variance:** 95% CI 8.9–12.48; p=0.0066
- **Toxicities:** any TEAE G≥3 (242/279, 87%); nausea G≥3 (n=279); vomiting G≥3 (n=279); decreased appetite G≥3 (n=279); treatment-related death G5 (5/279, 2%)
- **Case fit:** cross_tumor_only
- **Notes:** One of the two registrational gastric/GEJ trials behind the zolbetuximab approval and the source of the ≥75% CDx cutoff. Cross-tumor mechanism precedent only; GLEAM shows the effect did not carry into PDAC.

Shah/Xu (2023) — Nat Med

Reference: [PMID 37524953](#) · Type: clinical · Inclusion: included

- **Intervention:** zolbetuximab (anti-CLDN18.2 mAb) + CAPOX
- **Indication / model:** CLDN18.2-positive ($\geq 75\%$ 2+/3+), HER2-negative, untreated advanced gastric/GEJ adenocarcinoma (1L); 507 randomized 1:1, global.
- **Design:** Randomized double-blind placebo-controlled phase 3 (GLOW, NCT03653507)
- **n:** 507
- **Effect size:** 8.21 months — PFS
- **Variance:** $p=0.0007$
- **Toxicities:** any TEAE $G \geq 3$ (72.8%)
- **Case fit:** cross_tumor_only
- **Notes:** Second registrational gastric/GEJ trial. Same role in this dossier as SPOTLIGHT: validates the target at high expression in another GI tumor, with GLEAM as the PDAC counterweight.

Qi/Shen (2023) — J Hematol Oncol

Reference: [PMID 37689733](#) · Type: clinical · Inclusion: included

- **Intervention:** satri-cel (CT041, autologous CLDN18.2 CAR-T)
- **Indication / model:** CLDN18.2-positive metastatic PDAC after failure of standard therapy (2L+); Two patients: case 1 CLDN18.2 2+ in 70%; case 2 3+ in 60% — case 2 is nearly this patient's exact expression profile.
- **Design:** Two-patient research letter (patients treated within NCT04581473 and NCT03874897)
- **n:** 2
- **Effect size:** PR (case 1); target-lesion CR of lung metastases (case 2) response — response (RECIST v1.1)
- **Toxicities:** CRS G1-2 (2/2, 100%)
- **Case fit:** strong
- **Notes:** The most closely expression-matched published efficacy signal in the whole CLDN18.2 file — and it is two patients. A case report stays a case report; it argues for a sponsor conversation, not for an efficacy estimate.

Wainberg/O'Reilly (2023) — Lancet

Reference: [PMID 37708904](#) · Type: clinical · Inclusion: included

- **Intervention:** NALIRIFOX (comparator context)
- **Indication / model:** Treatment-naïve metastatic PDAC (1L); 770 randomized; treatment-naïve metastatic disease.
- **Design:** Randomized open-label phase 3 (NAPOLI 3, NCT04083235)
- **n:** 770
- **Effect size:** 11.1 months — median OS
- **Variance:** 95% CI 10.0–12.1; $p=0.036$
- **Toxicities:** any TEAE $G \geq 3$ (322/370, 87%); treatment-related death G5 (6/370, 2%)
- **Case fit:** partial
- **Notes:** Comparator-context row. If restaging reads the recurrence as 1L metastatic disease, NALIRIFOX is the guideline option on paper — but it re-runs three components of the regimen this tumor just progressed through within 4 months, so the early-recurrence biology argues for class switch or trial instead.

Mahadevan/Kalluri (2023) — Cancer Cell

Reference: [PMID 37625401](#) · Type: preclinical · Inclusion: included

- **Intervention:** KRAS-pathway inhibition and immune remodeling in PDAC (MRTX1133 as tool compound)
- **Indication / model:** 16 KRAS G12D-driven PDAC mouse models plus human patient-derived organoids
- **Design:** Blocking mutant KRAS induces FAS on cancer cells and shifts the microenvironment — more CD8+ effectors, fewer myeloid cells, reprogrammed CAFs — so CD8+ T cells can kill through FAS/FASL.
- **Effect size:** strong — MRTX1133 regressed both established PanIN and advanced PDAC, and regression of advanced disease depended on CD8+ T cells. Checkpoint blockade synergized with it and extended survival.
- **Variance:** vs vehicle; CD8 depletion arm; checkpoint blockade alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The tool compound is G12D-selective, and whether pan-RAS(ON) inhibition produces the same immune switch in a G12R tumor with lower ERK output is untested. Still the strongest mechanistic argument that anything in the surfaced set could open an MSS, TMB-low pancreatic tumor to T cells.

Gao/Shen (2023) — Journal for ImmunoTherapy of Cancer

Reference: [PMID 37364935](#) · Type: preclinical · Inclusion: included

- **Intervention:** givastomig (TJ033721/ABL111, CLDN18.2x4-1BB bispecific)
- **Indication / model:** gastric PDX-derived tumor cells with human PBMC; humanized 4-1BB transgenic mice bearing CLDN18.2-expressing tumors; cynomolgus toxicology; multiplex IHC of 60 gastric tumors
- **Design:** 4-1BB agonism is conditional on CLDN18.2 engagement, so costimulation fires only where tumor antigen is present, avoiding the liver toxicity of systemic 4-1BB antibodies.
- **n:** n=60 gastric tumors for the multiplex IHC arm
- **Effect size:** moderate — The magnitude of T-cell activation tracked with CLDN18.2 level on PDX tumor cells while remaining detectable across a range of expression. In transgenic mice the antibody raised the intratumoral CD8/Treg ratio, controlled tumors, and produced memory on rechallenge; monkeys showed no systemic immune activation or hepatotoxicity.
- **Variance:** vs parental anti-4-1BB agonist antibody; CLDN18.2-negative tumor cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Gastric models throughout, with no pancreatic arm. Conditional costimulation still needs T cells inside the tumor, which is exactly what a desmoplastic MSS PDAC tends not to supply.

—/— (2023)

Reference: — · Type: trial · Inclusion: included

- **Intervention:** AZD5863 (CLDN18.2×CD3 T-cell-engaging bispecific)
- **Indication / model:** CLDN18.2+ advanced solid tumors: gastric/GEJ, PDAC, esophageal (2L+ (≥1 prior line in the advanced/metastatic setting)); CLDN18.2+ by central IHC (cutoff protocol-defined)
- **Design:** Phase 1/2 dose escalation and expansion of AZD5863 monotherapy
- **Effect size:** — — Safety/DLT
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not

extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** XNW27011 (CLDN18.2 ADC)
- **Indication / model:** Advanced solid tumors after standard therapy, including PDAC and gastric cohorts with CLDN18.2 expression (2L+); CLDN18.2 expression confirmed on tumor tissue (expansion cohorts)
- **Design:** Phase 1/2 dose escalation and expansion of XNW27011, Q3W
- **Effect size:** — — MTD/RP2D; ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** AZD0901 / CMG901 (CLDN18.2 ADC, MMAE payload)
- **Indication / model:** CLDN18.2-expressing advanced solid tumours: PDAC, gastric/GEJ, biliary tract (1L/2L+ (max 2 prior lines for advanced disease; 1L cohort requires ≥6 months from adjuvant chemotherapy)); CLDN18.2-positive (sponsor assay); archival sample or fresh biopsy
- **Design:** Phase 2 multi-cohort, AZD0901 (CMG901) monotherapy and chemo combinations
- **Effect size:** — — ORR; safety
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** spevatamig (PT886, CLDN18.2×CD47 bispecific)
- **Indication / model:** Gastric/GEJ adenocarcinoma, PDAC, and biliary tract cancer (2L+ (escalation); PDAC combination substudy with gemcitabine/nab-paclitaxel); CLDN18.2 expression assessed on FFPE (fresh biopsy preferred, archival accepted); no cutoff stated in registry
- **Design:** Phase 1/2, spevatamig monotherapy and chemo/checkpoint combinations including a PDAC + GnP substudy
- **Effect size:** — — Safety; ORR; RP2D
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** LB1908 (CLDN18.2 CAR-T)
- **Indication / model:** CLDN18.2+ unresectable/metastatic gastric, GEJ, esophageal, or pancreatic cancer (2L+ (adjuvant/neoadjuvant counts as a line if progression within 6 months of completion)); CLDN18.2-positive per sponsor assay
- **Design:** Phase 1 dose escalation/expansion of LB1908 autologous CLDN18.2 CAR-T
- **Effect size:** — — Safety; RDE/RP2D
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TAC01-CLDN18.2 (T-cell antigen coupler engineered T cells; non-CAR platform)
- **Indication / model:** CLDN18.2+ metastatic solid tumors (2L+ (≥2 prior lines phase 1; 2-4 prior lines phase 2)); CLDN18.2-positive by central laboratory
- **Design:** Phase 1/2 (TACTIC-3), TAC01-CLDN18.2 autologous T-cell antigen coupler cells after lymphodepletion
- **Effect size:** — — Safety; RP2D
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** EO-3021 (SYSA1801, CLDN18.2 ADC, MMAE payload)
- **Indication / model:** CLDN18.2-expressing advanced gastric/GEJ cancer (≥20% of cells at 2+/3+) (2L+); CLDN18.2 ≥20% of tumor cells at IHC 2+/3+
- **Design:** Phase 1 dose escalation/expansion of EO-3021 monotherapy
- **Effect size:** ORR 22.2% in evaluable gastric/GEJ expansion — Safety; ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** YL-17231 (pan-RAS inhibitor)
- **Indication / model:** Advanced solid tumors with KRAS, HRAS, or NRAS mutations (2L+); Any KRAS/HRAS/NRAS mutation by local testing — G12R qualifies
- **Design:** Phase 1 dose escalation of YL-17231, oral pan-RAS inhibitor
- **Effect size:** — — Safety/DLT
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ELI-002 7P (7-peptide mKRAS amphiphile vaccine)
- **Indication / model:** KRAS/NRAS-mutated PDAC and CRC in minimal-residual-disease state after locoregional therapy (AMPLIFY-7P) (adj (post-definitive-therapy MRD window)); KRAS/NRAS G12D/G12R/G12V/G12A/G12C/G12S/G13D — G12R in the 7-peptide pool; requires ctDNA positivity or rising tumor markers WITH a negative CT for recurrence
- **Design:** Phase 1/2 of ELI-002 7P lymph-node-targeted amphiphile mKRAS vaccine
- **Effect size:** — — Safety; ctDNA/biomarker response; DFS
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Leidner/Tran (2022) — N Engl J Med

Reference: [PMID 35648703](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** mutant-KRAS TCR-T (G12D-restricted; documents the G12R coverage gap)
- **Indication / model:** Metastatic PDAC with KRAS G12D and HLA-C08:02 (2L+); *One patient; two allogeneic HLA-C08:02-restricted TCRs against KRAS G12D. Neither the mutation nor the HLA restriction transfers to a G12R tumor.*
- **Design:** Single-patient case report (NCI-surgery-branch-style TCR gene therapy)
- **n:** 1
- **Effect size:** PR, -72% target-lesion regression, ongoing at 6 months response — objective regression (RECIST PR)
- **Toxicities:** Abstract reports no treatment-limiting toxicity; per-event detail is in the full text (PMC9531755), not extracted for this bounding row.
- **Case fit:** none
- **Notes:** Kept to bound the coverage gap the trial screen documented: mKRAS TCR-T works in PDAC, but every clinical construct (this one, NT-112, Anocca ANOC-001/-002/-003, TCR1188-ABC) is G12D- or G12V-restricted. No G12R-restricted TCR-T and no G12R-selective inhibitor exists in the clinic as of August 2026. Toxicity extraction skipped as out of scope for a non-eligible modality; abstract-only review, PMC full text available.

Golan/Kindler (2022) — J Clin Oncol

Reference: [PMID 35834777](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** olaparib maintenance, germline BRCA-mutated mPDAC (gated on pending germline panel)
- **Indication / model:** Germline BRCA1/2-mutated metastatic PDAC, non-progressing on ≥ 16 weeks of first-line platinum (maintenance); Same 154-patient POLO population; final OS analysis.
- **Design:** Randomized double-blind placebo-controlled phase 3 (POLO), final OS analysis
- **n:** 154
- **Effect size:** 19.0 months — OS (final)
- **Variance:** $p=0.3487$
- **Toxicities:** Long-term tolerability held up; no new safety signals versus the primary analysis.
- **Case fit:** weak

- **Notes:** Final OS was null (medians identical; curves separate late with a tripled 3-year survival rate), so the honest framing is chemotherapy-free time and a long-survivor tail rather than a survival gain. Abstract reports no per-term AE table; primary-publication safety row carries the class rates.

Qi/Shen (2022) — Nat Med

Reference: [PMID 35534566](#) · **Type:** clinical · **Inclusion:** excluded — superseded by the final results of the same trial (CT041-CG4006, PMID 38830992)

- **Intervention:** satri-cel (CT041, autologous CLDN18.2 CAR-T)
- **Notes:** Excluded: superseded by the final results of the same trial (CT041-CG4006, PMID 38830992) — Interim report of CT041-CG4006; all estimates updated in the 2024 final publication.

Zhang/Shokat (2022) — Journal of the American Chemical Society

Reference: [PMID 36001446](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** KRAS G12R-selective covalent chemistry (no clinical agent exists)
- **Indication / model:** recombinant K-Ras(G12R) protein; X-ray crystallography of the ligand-protein adduct
- **Design:** alpha,beta-diketoamide electrophiles bind the switch II pocket and condense irreversibly with the epsilon- and eta-nitrogens of arginine 12, forming an imidazolium adduct.
- **Effect size:** Allele-selective covalent ligation of G12R was achieved and confirmed structurally, showing that a weakly nucleophilic arginine can be targeted at all. No cellular potency, pathway-suppression, or animal data accompany the chemistry.
- **Variance:** translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Chemistry proof of concept only. Four years on, no G12R-selective inhibitor has entered the clinic, which is the honest bound on this route for this patient.

Hwang/Regev (2022) — Nature Genetics

Reference: [PMID 35902743](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Treatment-enriched malignant programs after neoadjuvant therapy in PDAC
- **Indication / model:** 43 primary human PDAC specimens, treatment-naive or post-neoadjuvant, profiled by single-nucleus RNA-seq and whole-transcriptome spatial profiling
- **Design:** Chemotherapy and radiation shift the malignant compartment toward a neural-like progenitor program and reorganize spatial multicellular communities, rather than simply selecting a pre-existing subtype.
- **n:** n=43 primary tumors
- **Effect size:** moderate — The neural-like progenitor program was enriched after chemotherapy and radiotherapy and carried poor prognosis in independent cohorts. Three spatial communities emerged — classical, squamoid-basaloid, and treatment-enriched.
- **Variance:** vs treatment-naive specimens; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Descriptive human-tissue profiling with no therapeutic arm. It offers plausible biology for aggressive recurrence months after FOLFIRINOX but names no drug, and the neural-like program has no targeted agent.

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ASP2138 (CLDN18.2×CD3 2+1 bispecific T-cell engager)
- **Indication / model:** CLDN18.2+ gastric/GEJ adenocarcinoma and metastatic pancreatic adenocarcinoma (2L+ (PDAC escalation: no line limit; expansion ≤2 prior lines; a 1L chemo-combination cohort also exists)); CLDN18.2+ by central IHC; monotherapy escalation cutoff ≥1+ membranous staining
- **Design:** Phase 1/1b dose escalation and expansion; ASP2138 monotherapy (SC/IV) and combinations with chemotherapy or pembrolizumab
- **Effect size:** — — Safety/DLT; ORR
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IBI343 (CLDN18.2 ADC, topoisomerase-1 payload)
- **Indication / model:** CLDN18.2+ advanced solid tumors including PDAC and gastric/GEJ; monotherapy and chemo combinations (1L/2L+ (cohort-specific)); CLDN18.2+ by central pathology; expression tier required varies by cohort (dose-optimization cohorts preferentially enroll moderate-to-high expressers)
- **Design:** First-in-human phase 1 dose escalation/expansion/optimization of IBI343, including PDAC cohorts and FOLFIRINOX/mFOLFOX combinations
- **Effect size:** — — Safety/DLT; RP2D
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TORL-2-307-ADC (CLDN18.2 ADC)
- **Indication / model:** Advanced cancer including gastric, GEJ, and pancreas cancer (2L+); CLDN18.2-directed; expression requirement per protocol (not stated in registry)
- **Design:** First-in-human phase 1 of TORL-2-307-ADC
- **Effect size:** — — Safety; MTD/RP2D
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Sánchez-Rivera/Jacks (2021) — PNAS

Reference: PMID 34074758 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** p53 restoration in a KRAS-driven tumor: apoptotic priming as the gate

- **Indication / model:** Kras-driven p53-restorable mouse lung adenocarcinoma and sarcoma lines versus lymphoma lines; in vitro and in vivo
- **Design:** Whether restored p53 kills or merely arrests is set by mitochondrial apoptotic priming — how much reserve anti-apoptotic capacity the cell holds before p53 comes back.
- **Effect size:** strong — KRAS-driven solid-tumor lines with low priming survived p53 restoration and arrested in the cell cycle, while high-priming lymphoma cells died. Forced BCL-2/BCL-XL expression converted lymphoma to arrest, and navitoclax or BAD overexpression converted the solid-tumor lines to apoptosis in vitro and in vivo.
- **Variance:** vs lymphoma lines (high priming); BCL-2/BCL-XL overexpression and deletion arms; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Genetic p53 restoration in lung and sarcoma models, not pharmacologic Y220C reactivation in PDAC. It is nonetheless the best mechanistic account available for why a KRAS-mutant tumor may arrest rather than die when p53 function returns, and it names BH3 mimetics as the in-principle rescue.

Belmontes/Nolan-Stevaux (2021) — Science Translational Medicine

Reference: [PMID 34433637](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CLDN18.2xCD3 engager class — T-cell-cold tumors and combination rescue
- **Indication / model:** syngeneic solid-tumor models in huCD3e transgenic mice treated with mouse-CLDN18.2, mouse-CD19 and human-EPCAM BiTE molecules
- **Design:** Redirected lysis depends on T cells already present in the tumor: efficacy comes from local CD8+ expansion rather than recruitment from the circulation, and CD4+ T cells antagonize the response.
- **Effect size:** strong — Pretreatment intratumoral T-cell density was the dominant determinant of BiTE response across models. In poorly infiltrated, immunotherapy-refractory tumors, adding checkpoint blockade or 4-1BB agonism restored activity.
- **Variance:** vs vehicle; CD8 and CD4 depletion arms; checkpoint blockade and 4-1BB agonism alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Syngeneic models, none of them pancreatic. It is the cleanest mechanistic reason to expect a CLDN18.2 engager to underperform in a T-cell-poor MSS pancreatic tumor, and it names the combinations that fixed it in mice.

Hobbs/Der (2020) — Cancer Discovery

Reference: [PMID 31649109](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** KRAS G12R allele biology (target-feature mechanism; not a single agent)
- **Indication / model:** human PDAC cell lines carrying KRAS G12R vs G12D/G12V; biochemical p110alpha binding and macropinocytosis assays
- **Design:** Switch II is structurally perturbed in G12R, so the mutant binds the p110alpha catalytic subunit of PI3K poorly and cannot drive PI3K-dependent macropinocytosis the way G12D and G12V do; PDAC cells compensate with KRAS-independent PI3Kgamma acting through RAC1.
- **Effect size:** moderate — G12R lines did not run KRAS-driven macropinocytosis and did not require MYC for it, unlike G12D/V lines. The compensating nutrient-uptake route ran through PI3Kgamma rather than KRAS. Drug-sensitivity profiles separated G12R from G12D lines, though both stayed responsive to combined ERK

and autophagy inhibition.

- **Variance:** vs isogenic and allele-matched KRAS G12D/G12V PDAC lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Cell-line and biochemical work with no in-vivo arm. Nothing here tests whether a pan-RAS(ON) inhibitor engages G12R with the same potency it engages G12D.

Majzner/Mackall (2020) — Cancer Discovery

Reference: [PMID 32193224](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Antigen-density threshold as a CAR design parameter
- **Indication / model:** CD19- and GD2-directed CAR constructs tested against antigen-titrated tumor lines in vitro and in vivo
- **Design:** The antigen-density threshold for CAR activity is set by signal strength — ITAM count, hinge/transmembrane choice, costimulatory domain — rather than by the binder alone.
- **Effect size:** strong — CD28z CARs outperformed 4-1BBz CARs against antigen-low targets. Adding ITAMs or swapping a CD28 hinge/transmembrane into a 4-1BBz CAR lowered the recognition threshold while keeping the persistence advantage.
- **Variance:** vs parental CD19-4-1BBz and CD19-CD28z constructs; ITAM-deleted variants; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** CD19 and GD2, not CLDN18.2, and no solid-tumor stroma. Included because construct design rather than the IHC percentage alone decides whether 2+/3+ staining in 60% of cells is enough antigen for a given product.

—/— (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** satri-cel (CT041, CLDN18.2 CAR-T)
- **Indication / model:** CLDN18.2+ gastric and pancreatic cancer (North America) (2L+ (PDAC: ≥1 prior line)); CLDN18.2+ by sponsor IHC assay; satri-cel program threshold ≥2+ intensity in ≥40% of cells — the patient's 60% at 2+/3+ clears it
- **Design:** Phase 1b/2 open-label, satri-cel (CT041) autologous CLDN18.2 CAR-T after lymphodepletion, up to 3 infusions
- **Effect size:** — — Safety; ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Golan/Kindler (2019) — N Engl J Med

Reference: [PMID 31157963](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** olaparib maintenance, germline BRCA-mutated mPDAC (gated on pending germline panel)
- **Indication / model:** Germline BRCA1/2-mutated metastatic PDAC, non-progressing on ≥16 weeks of first-line platinum (maintenance); 154 randomized 3:2 of 3315 screened — the screening funnel itself shows how

uncommon the gBRCA gate is.

- **Design:** Randomized double-blind placebo-controlled phase 3 (POLO, NCT02184195)
- **n:** 154
- **Effect size:** 7.4 months — PFS (BICR)
- **Variance:** $p=0.004$
- **Toxicities:** any AE G $\geq$ 3 (n=91, 40%); AE leading to discontinuation (n=91, 5%)
- **Case fit:** weak
- **Notes:** Included because this patient's germline status is unknown — incidental tumor-NGS reporting found nothing, but no dedicated panel has been run, and a positive gBRCA/PALB2 result would open this maintenance route after platinum response. The row is inert until the germline panel returns.

Groot/He (2019) — Ann Surg

Reference: [PMID 31082915](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Prognostic context: early recurrence after PDAC resection
- **Indication / model:** Resected PDAC (upfront surgery era; neoadjuvant-treated patients excluded) (any); 957 pancreatectomy patients, 2000-2013, Johns Hopkins; recurrence in 753.
- **Design:** Retrospective single-center cohort with minimum-P-value cutpoint analysis
- **n:** 957
- **Effect size:** 1-year post-recurrence survival 20% (early) vs 45% (late) % — post-recurrence survival by recurrence timing
- **Variance:** $p<0.001$
- **Case fit:** partial
- **Notes:** Prognostic-framing row: this patient recurred about 7 months after resection and 4 months after adjuvant FOLFIRINOX, inside the 12-month early-recurrence window that carries 20% 1-year post-recurrence survival in this cohort. Two transfer caveats: the cohort excluded neoadjuvant-treated patients, and adjuvant therapy there was gemcitabine-era. The implication for treatment choice — treat as systemic, FOLFIRINOX-resistant disease and avoid re-challenge — comes from this biology plus the AGEO class-switch data, since no randomized re-challenge-versus-switch trial exists in this window. — (toxicity table not extracted)

Kinsey/McMahon (2019) — Nature Medicine

Reference: [PMID 30833748](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ERK plus autophagy inhibition (allele-linked vulnerability; no surfaced trial)
- **Indication / model:** PDAC cell lines and patient-derived xenografts; NRAS-mutant melanoma and BRAF-mutant CRC PDX for breadth
- **Design:** MEK1/2 blockade activates the LKB1-AMPK-ULK1 axis and the resulting autophagy buffers cells against loss of KRAS pathway output; blocking autophagy removes the buffer.
- **Effect size:** strong — Trametinib plus chloroquine was synergistic in vitro and regressed patient-derived PDAC xenografts where neither agent alone did. A patient treated with trametinib plus hydroxychloroquine had a partial response.
- **Variance:** vs single-agent trametinib and single-agent chloroquine; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** No G12R-specific arm; the allele link comes from Hobbs 2020, which reports G12R lines retain sensitivity to combined ERK/autophagy inhibition. No trial surfaced for this case uses this combination.

Jiang/Li (2019) — Journal of the National Cancer Institute

Reference: [PMID 30203099](#) · Type: preclinical · Inclusion: included

- **Intervention:** satri-cel (CT041) — the hu8E5-2I CLDN18.2 CAR-T construct
- **Indication / model:** gastric cancer cell-line xenografts and CLDN18.2-positive PDX in mice
- **Design:** Second-generation CD28-costimulated CAR built on the humanized hu8E5-2I scFv — the same binder that underlies CT041/satri-cel.
- **n:** n>=6 mice per group
- **Effect size:** strong — hu8E5-2I-28Z CAR-T suppressed xenograft growth (mean 75.5 mm³ versus 731.8 mm³ for untransduced T cells at day 29). Partial or complete tumor elimination occurred in CLDN18.2-positive PDX, with cells persisting and infiltrating tumor tissue.
- **Variance:** vs untransduced T cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Gastric models only. Mouse safety looked clean despite lysis of murine-CLDN18.2 targets, which did not predict the gastric mucosal injury later reported in patients.

—/— (2019)

Reference: — · Type: trial · Inclusion: included

- **Intervention:** zolbetuximab + gemcitabine + nab-paclitaxel
- **Indication / model:** CLDN18.2-positive (≥75% of cells 2+/3+) metastatic pancreatic adenocarcinoma, first-line (1L); CLDN18.2 ≥75% of tumor cells at moderate-to-strong membranous staining by central IHC — the patient's 60% fails this cutoff
- **Design:** Randomized phase 2 (GLEAM), zolbetuximab + gemcitabine/nab-paclitaxel vs GnP alone; n=393 enrolled
- **Effect size:** primary OS endpoint not met (topline, October 2025) — OS
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Tiriac/Tuveson (2018) — Cancer Discovery

Reference: [PMID 29853643](#) · Type: preclinical · Inclusion: included

- **Intervention:** Chemosensitivity is agent-specific in PDAC organoids (bearing on FOLFIRINOX-to-gemcitabine cross-resistance)
- **Indication / model:** patient-derived PDAC organoid library with paired genomic, transcriptomic and drug profiling; matched patient outcomes
- **Design:** Response is organoid-intrinsic and drug-specific, with distinct transcriptomic signatures for 5-FU, irinotecan, oxaliplatin, gemcitabine and paclitaxel rather than one general chemosensitivity axis.
- **Effect size:** moderate — Organoid drug profiles paralleled the matched patients' clinical courses, including longitudinal and synchronous-metastasis sampling. Per-agent signatures predicted response in independent adjuvant and advanced cohorts, and chemorefractory organoids were used to nominate targeted alternatives.
- **Variance:** vs untreated organoids; cross-agent comparison within the same organoid; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial

- **Notes:** Separate per-drug signatures argue against assuming that FOLFIRINOX failure predicts gemcitabine/nab-paclitaxel failure. The analysis is correlative and organoid pharmacotyping has never been validated prospectively for treatment selection.

Wang-Gillam/Chen (2016) — Lancet

Reference:PMID 26615328 · **Type:** clinical · **Inclusion:** included

- **Intervention:** liposomal irinotecan + 5-FU/LV (comparator context)
- **Indication / model:** Metastatic PDAC after gemcitabine-based therapy (2L); 417 randomized to nal-IRI+5-FU/LV, nal-IRI mono, or 5-FU/LV; population was gemcitabine-pretreated, not FOLFIRINOX-pretreated.
- **Design:** Randomized open-label phase 3 (NAPOLI-1, NCT01494506)
- **n:** 417
- **Effect size:** 6.1 months — median OS (combination arm)
- **Variance:** 95% CI 4.8–8.9; p=0.012
- **Toxicities:** neutropenia G3 (32/117, 27%); fatigue G3 (16/117, 14%); diarrhea G3 (15/117, 13%); vomiting G3 (13/117, 11%)
- **Case fit:** partial
- **Notes:** Comparator-context row. The approval basis for 2L nal-IRI is post-gemcitabine disease; after 10 cycles of irinotecan-containing FOLFIRINOX, re-exposing this patient to an irinotecan payload has no supporting data from this trial's population.

Ogitani/Agatsuma (2016) — Cancer Science

Reference:PMID 27166974 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** IBI343 (CLDN18.2 ADC, exatecan payload) — bystander-payload rationale, by class analogy
- **Indication / model:** HER2-positive/HER2-negative cocultures; mixed NCI-N87 + MDA-MB-468-Luc xenografts imaged in vivo; contralateral-tumor control
- **Design:** A membrane-permeable topoisomerase-I payload (DXd, an exatecan derivative) diffuses out of the antigen-positive cell and kills antigen-negative neighbours; the low-permeability DM1 payload does not.
- **Effect size:** strong — DS-8201a killed both antigen-positive and antigen-negative cells in coculture and suppressed the antigen-negative population in mixed xenografts, while T-DM1 did not. The effect stayed local — a contralateral antigen-negative tumor was unaffected.
- **Variance:** vs T-DM1 and an anti-HER2 ADC bearing a low-permeability payload; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** HER2 in breast and gastric models, not CLDN18.2 in pancreas. It is the mechanistic reason an exatecan-payload CLDN18.2 ADC could act on a tumor staining in 60% of cells, but the extrapolation crosses both target and tumor type and should not be read as tier-matched proof.

Jiang/DeNardo (2016) — Nature Medicine

Reference:PMID 27376576 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** FAK-driven fibrosis and immunosuppression as the gate on checkpoint response in PDAC
- **Indication / model:** p48-Cre;LSL-KrasG12D;Trp53flox/+ (KPC) mice treated with the FAK inhibitor VS-4718;

human PDAC tissue for the FAK-fibrosis correlation

- **Design:** Hyperactive FAK in neoplastic cells sustains the fibrotic, myeloid-rich microenvironment that keeps CD8+ T cells out.
- **Effect size:** strong — FAK activity in human PDAC tracked with fibrosis and poor CD8+ infiltration. VS-4718 alone limited progression and doubled survival in KPC mice while cutting fibrosis and immunosuppressive infiltrate, and it made previously refractory tumors responsive to PD-1 blockade and T-cell therapy.
- **Variance:** vs vehicle; anti-PD-1 alone; adoptive T-cell therapy alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** KPC carries Kras G12D with mutant Trp53, so the co-mutation shape resembles this case but the allele does not. FAK inhibitors have not reproduced this effect in PDAC trials.

Portal/Taieb (2015) — Br J Cancer

Reference: [PMID 26372701](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** gemcitabine + nab-paclitaxel after FOLFIRINOX failure (comparator context)
- **Indication / model:** Metastatic PDAC after FOLFIRINOX failure (2L); 57 consecutive patients in 12 French centers, prospective; exactly the class-switch this patient faces.
- **Design:** Prospective multicenter cohort (AGEO)
- **n:** 57
- **Effect size:** 8.8 months — median OS
- **Variance:** 95% CI 6.2–9.7
- **Toxicities:** any AE G3 (n=57, 40%); neutropenia G3 (n=57, 12.5%); peripheral neuropathy G3 (n=57, 12.5%); asthenia G3 (n=57, 9%); thrombocytopenia G3 (n=57, 6.5%)
- **Case fit:** strong
- **Notes:** Comparator-context row at the launching request, not a feature-targeted option: this is the standard class-switch benchmark (ORR 17.5%, mOS 8.8 mo) that any experimental 2L choice must beat. It is also the regimen daraxonasib doubled OS against in RASolute 302's comparator mix.

—/— (2015)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** pembrolizumab
- **Indication / model:** MSI-H/dMMR or TMB-H (≥ 10 mut/Mb) advanced solid tumors (KEYNOTE-158 basket) (2L+); Requires MSI-H/dMMR or TMB ≥ 10 mut/Mb — this tumor is MSS with TMB 0.83: biomarker-excluded
- **Design:** Phase 2 multi-cohort basket of pembrolizumab monotherapy; basis of the tumor-agnostic MSI-H and TMB-H approvals
- **Effect size:** — — ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Wöll/Türeci (2014) — International Journal of Cancer

Reference: [PMID 23900716](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CLDN18.2 expression in pancreatic neoplasms, primaries and metastases
- **Indication / model:** human pancreatic specimens: 24 normal, 202 primary lesions, 84 metastases, 48 intra-individually matched pairs; standardized IHC scored by two raters
- **Design:** Descriptive IHC mapping of CLDN18.2 across pancreatic tumor types and their metastatic sites.
- **n:** n=202 primaries, n=84 metastases, n=48 matched pairs
- **Effect size:** 103 primary PDAC samples (59.2%) stained positive and 95 (54.6%) at intensity 2+ or above. Metastases were positive at comparable rates — 69.4% of nodal and 65.7% of liver lesions — so expression persisted at metastatic sites.
- **Variance:** translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Pre-dates the 43-14A companion assay and the 75%-of-cells cutoff, so positivity rates are not directly comparable to a CDx call. No treatment arm.

Jacobetz/Tuveson (2013) — Gut

Reference: [PMID 22466618](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Hyaluronan and macromolecular permeability in PDAC (bearing on antibodies, ADCs and engagers)
- **Indication / model:** genetically engineered mouse model of PDA treated with PEGPH20, assessed for perfusion, vascular permeability and intratumoral drug delivery
- **Design:** Depleting hyaluronan re-expands collapsed vessels and opens endothelial fenestrations and interendothelial gaps, so large molecules as well as small ones reach the tumor.
- **Effect size:** strong — PEGPH20 raised intratumoral delivery of doxorubicin and gemcitabine and produced a tumor-specific rise in macromolecular permeability. Combination with gemcitabine slowed tumor growth and prolonged survival over gemcitabine alone.
- **Variance:** vs vehicle; gemcitabine monotherapy; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The macromolecular-permeability result is the part that speaks to antibody, ADC and engager penetration rather than small molecules. Same clinical caveat as Provenzano: the delivery mechanism never converted into survival benefit for PEGPH20.

Feig/Fearon (2013) — PNAS

Reference: [PMID 24277834](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Why checkpoint blockade fails in PDAC: FAP+ CAF-derived CXCL12 and T-cell exclusion
- **Indication / model:** autochthonous KPC PDA mice; genetic depletion of FAP+ carcinoma-associated fibroblasts; AMD3100 plus anti-PD-L1
- **Design:** FAP-expressing CAFs secrete CXCL12, which coats cancer cells and keeps T cells out of tumor regions; blocking CXCR4 releases that exclusion.
- **Effect size:** strong — Cancer-specific CD8+ T cells were present yet checkpoint blockade did nothing until FAP+ stromal cells were depleted. AMD3100 caused rapid T-cell accumulation among cancer cells and acted synergistically with anti-PD-L1, leaving mainly premalignant epithelium behind.
- **Variance:** vs anti-CTLA-4 and anti-PD-L1 alone; untreated tumor-bearing mice; translatability: med
- **Toxicities:** n/a (preclinical)

- **Case fit:** partial
- **Notes:** The mechanistic case that PDAC's checkpoint failure is exclusion, not absent antigen. CXCR4 combinations have not converted this into clinical benefit, and nothing surfaced for this case attacks the CXCL12 axis.

Provenzano/Hingorani (2012) — Cancer Cell

Reference:PMID 22439937 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Desmoplastic barrier to drug delivery in PDAC (hyaluronan/IFP mechanism)
- **Indication / model:** autochthonous KPC mouse PDA with direct interstitial fluid pressure measurement; PEGPH20 plus gemcitabine
- **Design:** Hyaluronan generates gel-fluid pressure that collapses tumor vessels and obstructs perfusion, diffusion and convection of therapeutics; enzymatic HA removal normalizes pressure and re-expands the microvasculature.
- **Effect size:** strong — Interstitial fluid pressures in PDA exceeded prior measurements and theoretical estimates for solid tumors. PEGPH20 with gemcitabine remodeled the microenvironment durably, produced consistent objective responses, and nearly doubled overall survival versus gemcitabine alone.
- **Variance:** vs gemcitabine monotherapy; vehicle; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The barrier mechanism is solid; the therapeutic translation failed, since PEGPH20 did not improve survival in the phase 3 HALO-301. Delivery is a real problem, but enzymatic HA depletion was not the answer.

Sahin/Türeci (2008) — Clinical Cancer Research

Reference:PMID 19047087 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CLDN18.2 as a target: lineage restriction and ectopic activation
- **Indication / model:** normal and tumor tissue panels across sites; CLDN18.2-specific monoclonal antibodies tested on cancer-cell surfaces
- **Design:** CLDN18.2 is a gastric-lineage tight-junction isoform whose expression depends on CREB binding an unmethylated consensus site; in normal tissue it is confined to differentiated gastric mucosa and spares the stem-cell zone.
- **Effect size:** Ectopic CLDN18.2 activation was found in pancreatic, esophageal, ovarian and lung tumors. Antibodies were raised that bind the isoform on cancer-cell surfaces without cross-reacting with the lung splice variant CLDN18.1.
- **Variance:** translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Target-discovery paper rather than an efficacy study. The normal gastric expression it documents is the on-target off-tumor liability every modality in this file carries.

Halaschek-Wiener/Jansen (2004) — Cellular Signalling

Reference:PMID 15337531 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** RAS inhibition raising p53 output (rationale for pairing pan-RAS with a Y220C reactivator)

- **Indication / model:** human colon carcinoma cell lines treated with a Ras inhibitor
- **Design:** Cutting Ras signalling raises p53 transcription and lowers Mdm2, so p53 protein escapes degradation and gains activity through two cooperating routes.
- **Effect size:** moderate — Ras inhibition increased p53 transcriptional activation and reduced Mdm2 in colon carcinoma cells, with the two effects additive on p53 function.
- **Variance:** vs untreated cells; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Twenty-year-old work in colon lines with wild-type p53 and pre-tri-complex Ras chemistry. It gives a direction for pairing RAS(ON) inhibition with Y220C reactivation, not evidence for that combination.